

SDE Intern Assignment

Note: The solution must be implemented in **JavaScript** or **Python**.

Topic

System Design (Low-Level Design)

Question 1: User Payout Management System

Design a comprehensive **Low-Level Design (LLD)** for a system that manages user payouts for affiliate sales.

Problem Statement

Every sale initially enters the system with the status **Pending**.

The system should provide users with an **Advance Payout** equal to **10% of the earnings** for all eligible pending sales.

Later, an administrator reconciles the sales, updating each sale to one of the following statuses:

- **Approved**
- **Rejected**

After reconciliation, the system should calculate the user's **final payout** by considering any advance payouts that have already been transferred.

Business Rules

1. Advance Payout

- Every **Pending** sale is eligible for an advance payout of **10% of its earnings**.
- Once an advance payout has been **successfully transferred**, the same sale **must never receive another advance payout**, even if the advance payout job runs multiple times.

2. Final Payout Calculation

During reconciliation:

Case 1 — Approved Sale

If a sale was:

- Pending
- Earnings = ₹30

- Advance paid = ₹3

After reconciliation, if the sale becomes **approved**, the remaining payout should be: ₹30 - ₹3 = ₹27

Case 2 — Rejected Sale

If a sale was:

- Pending
- Earnings = ₹50
- Advance paid = ₹5

After reconciliation, if the sale becomes **rejected**, the user has already received ₹5 that they were not entitled to.

Therefore, this amount must be adjusted against the user's final payout.

Adjustment = -₹5

3. Withdrawal Restrictions

A user can make **only one payout withdrawal every 24 hours**.

Question 2: Failed Payout Recovery

Sometimes a payout initiated to the user may later be:

- Cancelled
- Rejected
- Failed

In such cases, the system should:

- Credit the failed payout amount back into the user's withdrawable balance.
- Allow the user to initiate another withdrawal for that amount.

Reference Data

Sales Table

```
[
  {
    "userId": "john_doe",
    "brand": "brand_1",
    "status": "pending",
    "earning": 40
  }
]
```

Brands

brand_1
brand_2
brand_3

Possible Status Values

pending
approved
rejected

Terminology

Pending

The product has been purchased.

Approved

The product has been successfully delivered and the return period has ended.

Rejected

The purchased product was returned or cancelled.

Design Freedom

You are free to:

- Modify the reference schema as required.
- Introduce additional tables/collections.
- Design APIs.
- Define entities, relationships, indexes, and workflows.
- Explain trade-offs and design decisions.

Example

Before Reconciliation

```
[  
  {  
    "userId": "john_doe",  
    "brand": "brand_1",  
    "status": "pending",  
    "earning": 40  
  },  
  {
```

```
[
  {
    "userId": "john_doe",
    "brand": "brand_1",
    "status": "pending",
    "earning": 40
  },
  {
    "userId": "john_doe",
    "brand": "brand_1",
    "status": "pending",
    "earning": 40
  }
]
```

Total Pending Earnings

₹120

Advance Payout

10% of ₹120 = ₹12

After Reconciliation

```
[
  {
    "userId": "john_doe",
    "brand": "brand_1",
    "status": "rejected",
    "earning": 40
  },
  {
    "userId": "john_doe",
    "brand": "brand_1",
    "status": "approved",
    "earning": 40
  },
  {
    "userId": "john_doe",
    "brand": "brand_1",
    "status": "approved",
    "earning": 40
  }
]
```

Final Payout Calculation

Sale	Earnings	Advance Paid	Final Adjustment
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Rejected	₹40	₹4	−₹4
Approved	₹40	₹4	₹36
Approved	₹40	₹4	₹36

Total Final Payout

$$-₹4 + ₹36 + ₹36 = ₹68$$

Therefore, the user receives:

Final Payout = ₹68

Expected Deliverables

The submission should include:

1. Low-Level Design (LLD).
2. Database schema(s) with relationships.
3. Class design (or equivalent design in your chosen language).
4. APIs/endpoints
5. Handling of edge cases and failure scenarios.
6. Working implementation in **JavaScript** or **Python**.
7. Explanation of key design decisions and trade-offs.

Submission

- Attach the source code in github repo. The repo link should be public and shared in response to the assignment.
- Relevant readme files and docs should be present in the repo itself.

All the best!